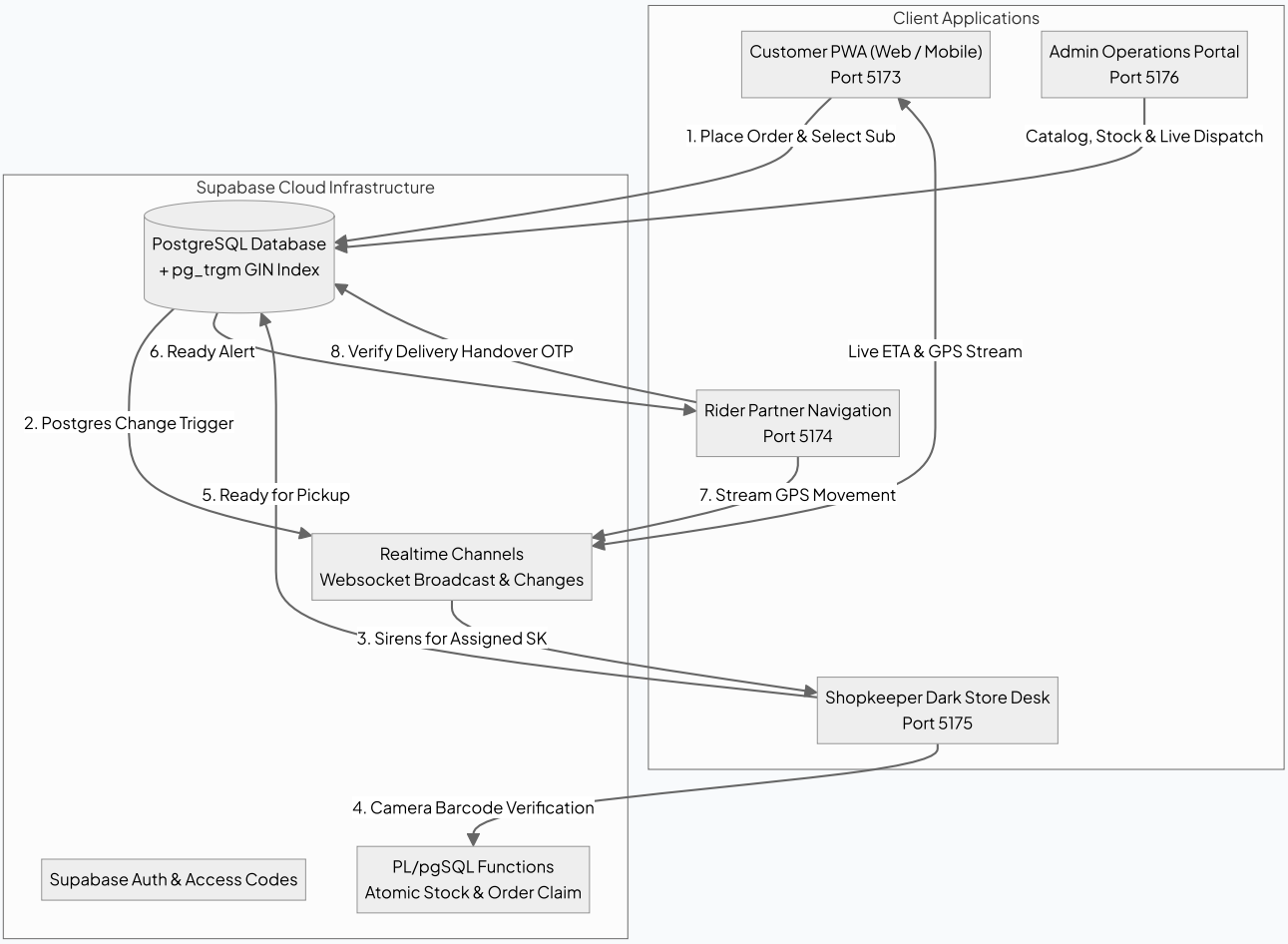
ZIPIT QUICK COMMERCE

10-MIN DELIVERY

System Architecture & Technical Ecosystem Specification · Hyper-Local Dark Store Pipeline

Version: 2.4.0 (Production)
Engine: React 18 + Supabase Cloud

🏛️ 1. High-Level Architecture & Event Pipeline



📱 2. The 4 Ecosystem Applications

🛍️ Customer App

PWA / Port 5173

- **Discovery & Catalog**: Fast category browser, hero banners, wafer tags, and Veg/Non-Veg filters.
- **Smart Search**: Natural language & regional synonyms via trigram search tags.
- **Native Mobile Gestures**: `popstate` physical back button handling closes bottom sheets without exiting app.

🏪 Shopkeeper App

Fulfillment / Port 5175

- **Targeted Siren Bell**: Audio alerts only sound on screens of shopkeepers tagged to the order's products.
- **Live Order Board**: Categorized by delivery vs store pickup and time groupings.
- **Barcode Packing Verification**: Hardware-accelerated camera scanner with native Web

- **Cart & Bill Breakdown**: Flying-item cart animations, dynamic MRP & discount calculations, and tiered delivery fees.
- **Live Order Tracking**: Stage-by-stage status tracking with real-time map & rider distance display.

Audio chimes and vibration.

- **Zero-Mismatch Normalizer**: Normalizes EAN-13, UPC-A, and QR codes ignoring leading zeros.
- **Mandatory Verification Gate**: Prevents marking orders *Ready for Pickup* until all items are 100% verified.

🛵 Rider App

Logistics / Port 5174

- **Active Deliveries Queue**: Available orders with earnings, customer distance, and store coordinates.
- **Route Map Engine**: Leaflet mapping with memoized markers and auto-fitting store-to-customer bounds.
- **0-DB-Write GPS Streaming**: Realtime in-memory broadcast channel streams live location only on ≥ 8m movement.
- **Fallback Branding Coordinates**: Synchronous store hub coordinate resolution with local storage cache.
- **4-Digit Delivery OTP**: Cryptographic handover verification ensuring customer receives the package.

⚙️ Admin Portal

Operations / Port 5176

- **Catalog CRUD**: Pricing, MRP, unit weights, showcase slots (Pinnacle Grid, Bestsellers).
- **Camera Barcode Capture**: 1-tap camera scanner auto-fills product barcodes into catalog.
- **Shopkeeper Multi-Select Assignment**: Live dropdown pulls registered shopkeepers from `access_codes`.
- **Dispatch Control**: Real-time order desk with manual rider and shopkeeper reassignment overrides.

🏷️ 3. Embedded Metadata Tag Serialization Protocol

To prevent database schema breakage and client cache invalidation across apps, product metadata is serialized into the `name` string:

Tag Format	Description & Functional Usage	Live Database Example
`---BARCODE:...---`	Stores retail EAN-13 / UPC barcode for shopkeeper camera packing verification.	`---BARCODE:8906005274519---`
`---MRP:...---`	Maximum retail price used to calculate customer discount % OFF pills.	`---MRP:120---`
`---SHOPKEEPERS:...---`	Comma-separated list of shopkeeper codes authorized to fulfill the product.	`---SHOPKEEPERS:CRNOD9JN, SK02---`
`---TAGS:...---`	Synonyms and search terms used by customer text and voice search algorithm.	`---TAGS:doodh, dairy, morning, chai---`
`---STICKER:...---`	Visual highlight badge shown on product card corner.	`---STICKER:Exclusive 20% OFF---`
`---TAG:GRID---`	Designates product into the 10-slot Pinnacle Showcase on home screen.	`---TAG:GRID---`
`---TAG:BESTSELLER---`	Awards bestseller golden ribbon badge on catalog.	`---TAG:BESTSELLER---`
`---TAG:OOS---`	Hard stock override locking out-of-stock items immediately.	`---TAG:OOS---`

🔐 4. Database Security & PL/pgSQL Atomic Procedures

> **Security Enforced:** All database operations are secured with Row Level Security (RLS) policies. Unassigned shopkeepers cannot query or claim orders belonging to other stores.

Component	Implementation	Security Benefit
Atomic Claim RPC	`claim_shopkeeper_order(p_order_id, p_code)`	Row-level lock (`FOR UPDATE`) eliminates race conditions when multiple shopkeepers claim an order.
Search Indexing	`idx_products_name_trgm` (GIN Index)	Accelerates substring queries across product names and tags with millisecond response times.
Shopkeeper RLS Policy	`role = 'shopkeeper'` on `access_codes`	Prevents unauthorized reading of internal employee codes and admin credentials.

5. Egress & System Performance Benchmark

Operation	Architecture Mechanism	Bandwidth / Egress Impact
Camera Barcode Scanning	Device-side `BarcodeDetector` GPU evaluation	**0.00 KB** (Zero video frames sent to cloud)
Audio Chime Synthesizer	Browser native `AudioContext` sine waves	**0.00 KB** (Zero audio assets downloaded)
Rider GPS Live Broadcast	Supabase Realtime Websocket Broadcast Channel	**0 DB Writes** (Throttled ≥ 8m distance)
Store Hub Coordinates	`db.branding.get()` with LocalStorage cache	**0.00 KB** once cached in rider device